OC-GS: Supplementary Material

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Abstract

This supplementary document provides additional implementation details, controls, diagnostics, and extended analyses that support the main submission.

1 Appendix: Implementation Details and Supplementary Diagnostics

This appendix supplements the main paper with implementation details and selected diagnostics that were abbreviated for space. It mirrors Sections 2–6 and keeps only the details that materially affect reproducibility, rather than restating the main derivations.

All symbols follow the main text, including per-Gaussian semantic orientation $\hat{\mathbf{f}}_i$, saliency s_i, object descriptors $(\bar{\mathbf{f}}_m, \bar{s}_m)$, object-demand distribution $p_{\text{total}}(o \mid t)$, and runtime budgets $(B(t), C_{\max}(t))$.

1.1 Motivation Protocol Details (Supplementing Sec. 2)

This section clarifies how the evidence in Sec. 2 is produced and how the reported controls are applied.

Gaze-object association and concentration. Per-frame concentration is computed by Eq. (1) and Eq. (2) in the main text. The evidence chain combines two sources. First, we use prior psychophysics and VR-system findings as design constraints rather than re-running those user studies: the 100 ms post-saccade reference comes from gaze-contingent foveated rendering and streaming work [7, 8]; object-based attention evidence comes from same-object facilitation and related selection studies [1, 2, 10]; trans-saccadic continuity comes from studies showing preferential maintenance of object identity across gaze shifts [3, 9]; and fixation-instability evidence comes from microsaccade and drift measurements during nominal fixation [4–6]. These findings motivate object-level matching, binocular consistency checks, and temporally aligned reporting.

Second, for our own immersive-viewing measurements, we use synchronized binocular gaze and head pose from the cross-site 6DoF eye-tracking cohort described in Sec. 2. All streams are aligned to a unified frame clock, and frame-aligned renders are used for mask association. At each frame, a gaze-centered Gaussian kernel is projected to image space, instance masks are obtained and temporally linked by IoU-plus-descriptor matching with forward-backward consistency, and object mass shares $a_{t,n}$ are aggregated by kernel-mask overlap before top-K concentration $C_t(K)$ is computed. We then apply the same controls reported below and in Fig. 1, and report concentration together with ICC, binocular correlation, and bootstrap confidence intervals. Under this unified protocol, concentration remains clearly above uniform-reference levels, supporting the object-centric demand premise used by the later offline and online modules.

The four motivation panels in Fig. 2(a)–(d) are generated under one synchronized protocol so that their interpretations remain directly comparable. Panel (a) measures post-saccade recovery on the same event timeline for all compared methods. Panel (b) reports object-level concentration and concentration-to-uniform ratios under matched frame indices. Panel (c) measures collateral transfer under fixed bandwidth traces by counting transferred bytes outside the attended-object support when the target is correctly identified. Panel (d) computes revisit churn by Eq. (3) on the same runtime clock and visualizes refetch spikes together with FoV-quality oscillation.

Temporal mask consistency and filtering. Raw masks are linked over time using IoU-plus-descriptor matching with forward-backward consistency checks. Low-consistency tracks are removed before concentration reporting. Because filtering can bias concentration estimates toward stable large instances, we log mask-drop diagnostics by scene, object-scale bin, and occlusion-density bin, and also report reweighted concentration estimates using inverse keep-probability weighting.

Control settings referenced in Sec. 2. The concentration analysis uses four controls, each targeting a different confound.

C1: Area normalization. To remove pure size advantage, we reweight object mass by mask area:

$$a_{t,n}^{\text{area}} = \frac{a_{t,n} / \left(|M_{t,n}| + \epsilon\right)^{\eta}}{\sum_j a_{t,j} / \left(|M_{t,j}| + \epsilon\right)^{\eta}}, \quad \eta \in \{0.5, 1.0\}. \tag{1}$$

C2: Center-bias correction. To remove canonical center prior, we divide by empirical fixation prior $P_c(p)$:

$$a_{t,n}^{\text{cb}} = \frac{\sum_p \frac{G_t(p) M_{t,n}(p)}{P_c(p)+\epsilon}}{\sum_p \frac{G_t(p)}{P_c(p)+\epsilon}}. \tag{2}$$

C3: Saliency-only replacement. We replace object-instance identities with connected saliency components and recompute top-K concentration under the same gaze kernel, testing whether static saliency alone can explain the result.

C4: Mask-consistency sensitivity. We sweep temporal track-consistency threshold τ_{trk} (e.g., 0.3/0.5/0.7) and report inverse keep-probability weighting:

$$C_t^{\text{ipw}}(K) = \frac{\sum_{n \in \text{TopK}(a_t)} a_{t,n} / \pi_{t,n}}{\sum_j a_{t,j} / \pi_{t,j}}. \tag{3}$$

Here $\pi_{t,n}$ is the keep probability of object track n under the current consistency threshold. This correction checks whether selective dropping of unstable tracks would otherwise overestimate concentration.

Revisit churn trace generation. For Eq. (3), eviction and refetch events are recorded on the same runtime clock as request and residency updates. We align churn curves with saccade and revisit events so that demand volatility can be separated from policy-induced oscillation.

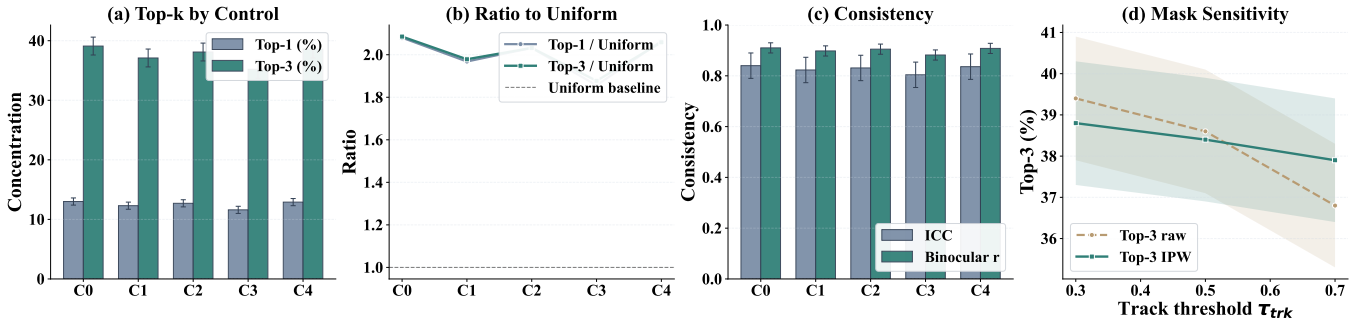

Figure 1: Motivation-control reproducibility figure. It reports four diagnostic views of the concentration claim: (a) Top-k concentration across controls C0–C4, (b) concentration-to-uniform ratios, (c) participant-level and binocular consistency, and (d) mask-threshold sensitivity with raw versus inverse keep-probability weighted estimates. All settings use matched sessions and frame indices.

1.2 Offline Stage Details (Supplementing Sec. 4)

1.2.1 Semantic–Saliency Co-Distillation: Full Procedure. The offline stage compiles frozen 3DGS into an object-addressable substrate instead of retraining scene geometry. Geometry and appearance remain frozen; only side-information fields are optimized, namely semantic orientation and saliency prior per Gaussian primitive. Teachers are SAM3-style instance masks in the SAM/SAM2 promptable paradigm, optional CLIP-style region descriptors, and VST++ saliency maps.

Because one rendered pixel usually mixes multiple Gaussians along a ray, supervision is imposed in rendered image space rather than at isolated primitive level. We use a shared semantic codebook $S \in \mathbb{R}^{L\times D}$ with sparse coefficients β_i:

$$\mathbf{f}_i = \beta_i^\top S, \tag{4}$$

then normalize $\mathbf{f}_i$ to semantic orientation $\hat{\mathbf{f}}_i$.

For pixel p in view v, semantic and saliency maps are rendered with the same visibility weights:

$$\begin{aligned} \mathbf{R}_v(p) &= \frac{\sum_{i=1}^{K_p} \omega_i^v(p)\hat{\mathbf{f}}_{g_i^p}}{\sum_{i=1}^{K_p} \omega_i^v(p) + \varepsilon}, \\ \hat{A}_v(p) &= \frac{\sum_{i=1}^{K_p} \omega_i^v(p) s_{g_i^p}}{\sum_{i=1}^{K_p} \omega_i^v(p) + \varepsilon}. \end{aligned} \tag{5}$$

Training uses uniform trajectory coverage plus hard-example supplementation, including high saliency entropy, dense instances, and rapid viewpoint transitions. The optimized objective is

$$\mathcal{L}_{\text{off}} = \lambda_{\text{sem}}\mathcal{L}_{\text{sem}} + \lambda_{\text{sod}}\mathcal{L}_{\text{sod}} + \lambda_{\text{sp}} \sum_i \|\beta_i\|_1 + \lambda_{\text{ent}}\mathcal{L}_{\text{ent}}, \tag{6}$$

where each term has a concrete role. The semantic term combines instance-consistent alignment and inter-instance separation in rendered space:

$$\mathcal{L}_{\text{sem}} = \frac{1}{|\Omega_v|} \sum_{p\in\Omega_v} \left(1 - \mathbf{R}_v(p)^\top \mathbf{t}_{c(p)}\right) + \eta \sum_{(p,q)\in\mathcal{P}_v^-} \left[\mathbf{R}_v(p)^\top \mathbf{R}_v(q) - m\right]_+, \tag{7}$$

where $c(p)$ is the SAM3 instance index at pixel p, $\mathbf{t}_{c(p)}$ is its teacher descriptor, and $\mathcal{P}_v^-$ are negative cross-instance pixel pairs. The saliency term aligns dense maps with edge awareness:

$$\mathcal{L}_{\text{sod}} = \text{BCE}(\hat{A}_v, A_v) + \gamma\|\nabla\hat{A}_v - \nabla A_v\|_1. \tag{8}$$

The ℓ_1 term on β_i enforces sparse code usage, and the anti-collapse term keeps codebook utilization balanced:

$$\mathcal{L}_{\text{ent}} = \sum_{\ell=1}^{L} \pi_\ell \log(\pi_\ell L + \epsilon), \qquad \pi_\ell = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}[\ell \in \text{TopK}(\beta_i)]. \tag{9}$$

Thus, $\mathcal{L}_{\text{sem}}$ enforces semantic discriminability, $\mathcal{L}_{\text{sod}}$ preserves perceptual-priority consistency, sparse coding controls payload, and $\mathcal{L}_{\text{ent}}$ prevents codebook collapse. We update only S, $\{\beta_i\}$, and $\{s_i\}$.

Codebook export and payload accounting. After co-distillation converges, we export hard top-K sparse assignments with quantized values. The side channel consists of one shared codebook plus sparse per-Gaussian index/value entries and saliency scalars. Using dense float32 features would cost $4ND$ bytes, whereas the sparse representation reduces this to 12.4 MB + 64 KB at the representation level. We then apply a conventional stream-oriented packing pass, in the spirit of traditional media-streaming metadata compression: repeated codebook references are shared globally, sparse tuples are serialized into compact index/value records, and quantized coefficients and saliency values are stored in a transport-friendly packed bitstream. This further reduces the payload to 6.2 MB + 64 KB, which is the transmitted side channel used in all experiments.

Practical hyperparameter defaults. For reproducibility, all reported runs use one frozen offline configuration rather than per-scene retuning. Codebook export uses semantic dimension $D = 64$, codebook size $L = 256$, hard top-K sparsity with $K = 4$, 16-bit sparse indices, 8-bit coefficient quantization, and 8-bit saliency quantization. For semantic-geometric clustering, we set the spatial-semantic balance parameters to $\rho = 0.3$ m and $\lambda = 0.5$. Queue routing then fixes $(\tau_{\text{obj}}, \tau_{\text{pack}}) = (0.50, 0.50)$ after validation and keeps the same operating point across all scenes and sites. Offline optimization uses AdamW with cosine decay and

short warm-up; loss weights and routing thresholds are selected on train/val once and are not retuned on the held-out split.

1.2.2 Hybrid Decoupling and Queue A/B/C Classification. Distilled signals are converted into runtime-consumable object units in three steps: reliability filtering, semantic-geometric clustering, and queue routing. Reliability filtering and clustering follow the same definitions as main Eq. (8) and Eq. (9); here we emphasize the routing logic rather than restating those formulas.

For each cluster $\mathcal{O}_m$, we compute object-worthiness $w_m = \bar{s}_m \kappa_m$ and a scale indicator $h_m = \max(\tilde{v}_m, \tilde{b}_m)$. Routing then follows a deterministic two-stage flow: clusters with $w_m < \tau_{\text{obj}}$ are sent to Queue A; among the remaining clusters, those with $h_m \geq \tau_{\text{pack}}$ are sent to Queue B, and the rest to Queue C. The first decision separates low-worth background structure from object channels; the second separates large objects that benefit from hierarchical refinement from compact objects that are better delivered atomically.

In this paper, Queue B is the large-object hierarchical channel, not a transition band. Queue C is reserved for compact objects suitable for atomic delivery. Boundary residues are not exported as a separate queue: stable contour residues are absorbed into Queue-B hierarchies, while unstable cross-object residues fall back to Queue A.

Object descriptors exported to the online predictor and scheduler include a saliency-weighted semantic prototype, $\bar{\mathbf{f}}_m = \frac{\sum_{i \in \mathcal{O}_m} s_i \hat{\mathbf{f}}_i}{\left\| \sum_{i \in \mathcal{O}_m} s_i \hat{\mathbf{f}}_i \right\|_2 + \varepsilon}$, and mean saliency, $\bar{s}_m = |\mathcal{O}_m|^{-1} \sum_{i \in \mathcal{O}_m} s_i$.

1.2.3 Packaging, Dependency Legality, and Runtime Interface. After routing, Queue A is packaged as macro octree LoD nodes, Queue B as object-anchored coarse-to-fine chains, and Queue C as atomic micro-packs. This mapping directly matches runtime behavior: Queue A supports broad recovery, Queue B supports progressive refinement for large objects, and Queue C supports low-latency one-shot delivery for compact objects. Exported metadata include object identity, semantic prototype, saliency prior, compact geometry bounds, dependency links, and per-block cost metadata.

Dependency legality is defined by $\mathcal{E} = \mathcal{E}_B^{\text{intra}} \cup \mathcal{E}_{A \to B}^{\text{cross}}$, where optional cross-queue prerequisites are allowed only from Queue-A macro roots to Queue-B roots under heavy occlusion. We do not introduce reverse $B \to A$ prerequisites or free-form cross-level edges. This keeps the dependency graph legal and scheduler-facing: no arbitrary deep-descendant access is allowed before required roots are available.

For reproducibility diagnostics, we report three scalar summaries: objectization coverage $\rho_{\text{obj}} = \frac{\sum_{i \in \mathcal{J}_B \cup \mathcal{J}_C} w_i}{\sum_{i \in \mathcal{J}_A \cup \mathcal{J}_B \cup \mathcal{J}_C} w_i}$, atomicity ratio $\rho_{\text{micro}} = \frac{|\mathcal{J}_C|}{|\mathcal{J}_B| + |\mathcal{J}_C|}$, and packaging overhead $\gamma_{\text{pack}} = \frac{\text{packaged bytes}}{\text{raw bytes}}$. These diagnostics answer three practical questions directly: whether object channels carry meaningful mass, whether compact objects remain atomic after routing, and whether metadata overhead stays controlled after packaging.

1.3 Online Stage Details (Supplementing Sec. 5)

1.3.1 Object-Anchored Predictor: Design and Why It Is Not a Conventional Predictor. The released implementation uses a three-part predictor stack, and all parts operate in the same object-action space consumed by the scheduler. Phase 2 is a motion-conditioned physical predictor, Phase 3 adds visual context fusion, and Phase 4 is a fixation-token intent model. The controller fuses geometry-driven demand and intent demand directly at object level, without remapping predicted objects back to spatial tiles.

For Phase 2 and Phase 3, each training sample uses a fixed motion window containing FoV state, head pose, gaze pose, head velocity, and frame interval; the supervision target is future head pose over the prediction horizon. Phase 2 first computes a deterministic physical baseline and then predicts residual position and quaternion corrections:

$$\begin{aligned} \mathbf{b}_{\tau+k}^{\text{pos}} &= \mathbf{p}_\tau^h + \mathbf{v}_\tau^h \sum_{j=1}^{k} \Delta t_{\tau+j}, \\ \mathbf{b}_{\tau+k}^{\text{quat}} &= \mathbf{q}_\tau^h, \\ \hat{\mathbf{y}}_{\tau+k} &= \left[\mathbf{b}_{\tau+k}^{\text{pos}} + \mathbf{r}_{\tau+k}^{\text{pos}}, \text{norm}\big(\mathbf{b}_{\tau+k}^{\text{quat}} + \mathbf{r}_{\tau+k}^{\text{quat}}\big) \right]. \end{aligned} \tag{10}$$

Phase 3 reuses the same baseline-plus-residual structure but injects visual context by cross-attention. A lightweight ResNet branch encodes depth, semantic, and saliency tensors, and motion features query visual tokens through multi-head attention. The fusion gate is conditioned on motion feature, context feature, speed, and turn indicator:

$$\alpha_t = \sigma\Big(g\Big([\mathbf{h}_t^{\text{mot}}, \mathbf{h}_t^{\text{ctx}}, \|\mathbf{v}_t^h\|, \text{isTurn}_t]\Big)\Big), \tag{11}$$

with an additional speed-based suppression term in implementation to down-weight visual correction during high-speed motion blur.

Phase 4 models intent as next-object prediction over fixation tokens. Frame-level hit-object events are converted to fixation sessions by run-length filtering, background and invalid-id removal, dwell and gap extraction, and optional viewport-bin labeling. A causal Transformer language model then predicts the next object token, and optionally the next viewport class, from token history plus dwell and gap embeddings.

Runtime fusion still follows main Eq. (12). The geometric branch provides $p_{\text{geom}}(o \mid t)$, the intent branch provides $p_{\text{intent}}(o \mid t)$, and the fused result is passed to the scheduler without any intermediate tile-space translation. Training minimizes a weighted sum of deterministic kinematics, object and viewport classification, and uncertainty regularization. In code, the kinematic term contains position and rotation error plus a smoothness term; the classification term contains object cross-entropy and optional viewport cross-entropy; and the uncertainty term contains heteroscedastic NLL plus variance regularization. Training uses user-disjoint and scene-disjoint splits, with $\Delta_p = 100$ ms and $\Delta_s = 100$ ms in all main experiments.

1.3.2 Code-Aligned End-to-End Workflow for Predictor Training and Runtime. To keep reproduction straightforward, the predictor is organized as staged dataflow rather than a single black-box model. The offline side first packs synchronized head-eye records

into fixed-length windows, then trains Phase 2 and Phase 3 trajectory prediction together with Phase 4 intent modeling on consistent object identifiers. The online side executes the same phase outputs in sequence and hands a unified object-space demand posterior to the scheduler.

Each packed training window stores motion history, future supervision, optional turn or mask metadata, and the last history frame id used for context lookup. This structure is consumed by the Phase 2 and Phase 3 dataloaders with identical horizon semantics, so residual learning and uncertainty learning share the same supervision target.

Phase 2 is optimized first as the motion backbone. Its practical training schedule is deterministic-first: uncertainty terms are disabled during warmup and then ramped in gradually,

$$\lambda_{\text{nll}}(e), \lambda_{\text{vr}}(e) = \begin{cases} 0, 0, & e < e_w, \\ \bar{\lambda}_{\text{nll}}\, r(e), \bar{\lambda}_{\text{vr}}\, r(e), & e \geq e_w, \end{cases} \qquad r(e) = \min\left(1, \frac{e - e_w + 1}{e_r}\right), \tag{12}$$

which stabilizes residual fitting before heteroscedastic terms become dominant.

In the released training configuration, Phase 2 uses AdamW for 30 epochs with batch size 32, learning rate 3×10^{-4}, weight decay 10^{-4}, warmup parameters $(e_w, e_r) = (2, 5)$, and ramp caps $(\bar{\lambda}_{\text{nll}}, \bar{\lambda}_{\text{vr}}) = (0.1, 10^{-3})$. Phase 3 uses AdamW for 20 epochs with batch size 8, learning rates 10^{-4} for pretrained motion parameters and 5×10^{-4} for newly introduced fusion parameters, weight decay 10^{-4}, and loss weights $(w_{\text{pos}}, w_{\text{quat}}, w_{\text{nll}}, w_{\text{varreg}}) = (1.0, 1.0, 0.1, 10^{-3})$. Phase 4 uses AdamW for 20 epochs with batch size 256, learning rate 3×10^{-4}, and weight decay 10^{-3}. These settings are fixed on train/val and then frozen for all reported held-out and cross-site evaluations.

Phase 3 then adds context fusion on top of the same motion stream. For each window, the loader retrieves depth, semantic, and saliency tensors by manifest index using the last frame id in the history window, and injects them through visual encoding and cross-attention into the motion query. This keeps geometric and semantic evidence frame-synchronous with the trajectory anchor used by baseline-plus-residual prediction.

Phase 4 is built from frame-level hit-object traces by fixation sessionization. After background removal, run-length filtering, and gap splitting, each retained token stores object identity together with binned dwell and inter-fixation gap statistics. The causal language model predicts the next object token, and optionally the next viewport class, under teacher forcing during training.

At runtime, the same interface is preserved. Phase 3 produces geometry-aware demand from current motion and fetched context, while Phase 4 produces intent demand from recent fixation history. Their fusion yields $p_{\text{total}}(o \mid t)$, and only this object-space posterior is passed to Lyapunov scheduling. The full path from data packing to control action therefore remains object-native end to end.

1.3.3 Lyapunov Scheduler Implementation, Eviction, and Anti-Thrashing Details. At each slot, runtime state includes current head-eye state, short history, resident set $\mathcal{M}(t)$, and budgets $(B(t), C_{\max}(t))$. Candidate generation merges Queue-B/C object frontiers with Queue-A macro coverage candidates. The actual request set is the selected non-resident subset of those candidates after legality and budget checks.

The per-slot objective follows the drift-plus-penalty formulation in main Eq. (21)–Eq. (22). In implementation we rank candidates by a practical surrogate score

$$\psi_k(t) = V\, \hat{g}_k(t) - Z_B(t)\, \text{size}_k - Z_C(t)\, \text{dec}_k - \xi\, \mathbb{I}[k \in \mathcal{R}_{\text{recent}}], \tag{13}$$

then perform a feasibility scan under main Eq. (17)–Eq. (18) and dependency legality. Complexity is $O(N \log N)$ per slot over the active frontier.

When overflow occurs, newly fetched units and dependency ancestors are pinned for one epoch. Eviction applies a retention score

$$\phi_k(t) = p_{\text{total}}(o(k) \mid t)\Delta Q_k(t) - \gamma_a\, \text{age}_k - \gamma_r\, \mathbb{I}[k \in \mathcal{R}_{\text{recent}}], \tag{14}$$

and removes lowest-retention non-pinned units first. In the reported runs, we set $(\gamma_a, \gamma_r) = (0.15, 0.30)$, use a 3-epoch recent window, and pin newly fetched units and dependency ancestors for 1 epoch (100 ms). The lightweight refetch penalty ξ in Eq. (13) is fixed once on validation and is not retuned on the held-out test split.

1.3.4 Real-Time Feasibility and Scope. The real-time claim applies to the *online control loop*, not to offline preparation. Offline distillation and packaging are one-time scene preprocessing steps, approximately 3 hours per scene in Table 1, and are amortized across users and sessions.

Online, the control-period budget is $\Delta_s = 100$ ms. Predictor latency is under 10 ms per step (Table 1), and the scheduler runs once per 100 ms epoch with active-set complexity $O(N \log N)$. In practice, real-time means $T_{\text{pred}} + T_{\text{sched}} + T_{\text{bookkeep}} < \Delta_s$, where T_{bookkeep} includes lightweight candidate maintenance and residency updates. Transport and decode continue under the separate bandwidth and decode constraints in main Eq. (17)–Eq. (18).

1.3.5 Evaluation Metric Details. The four online metrics used throughout the paper are computed on a synchronized event timeline shared across methods. Let I_t be the rendered RGB frame of a method, I_t^{gt} the matched ground-truth frame, and $w_t(p)$ the normalized gaze-centered foveation weight map at pixel p. Per-frame Fov-PSNR is computed from the weighted MSE,

$$\text{FovPSNR}_t = 10 \log_{10}\left(\frac{1}{\sum_p w_t(p) \| I_t(p) - I_t^{\text{gt}}(p) \|_2^2 + \epsilon}\right), \tag{15}$$

and reported values are averaged over the matched evaluation frames of the corresponding scene split.

Recovery latency is defined per gaze or saccade event. For an event beginning at time t_e, we first compute its settled attended quality $\bar{Q}_e^{\text{settle}}$ as the mean Fov-PSNR over the final scheduler epoch before the next event or the end of the valid post-event segment. Recovery latency is then the earliest post-event time at which the current Fov-PSNR reaches 95% of $\bar{Q}_e^{\text{settle}}$ and remains above that level for one scheduler epoch. Event-level latencies are averaged to produce the reported recovery time in milliseconds.

Refetch ratio is defined on transferred bytes rather than request counts. Let B_{re} denote the bytes fetched for units that had previously been resident, were later evicted, and were then fetched

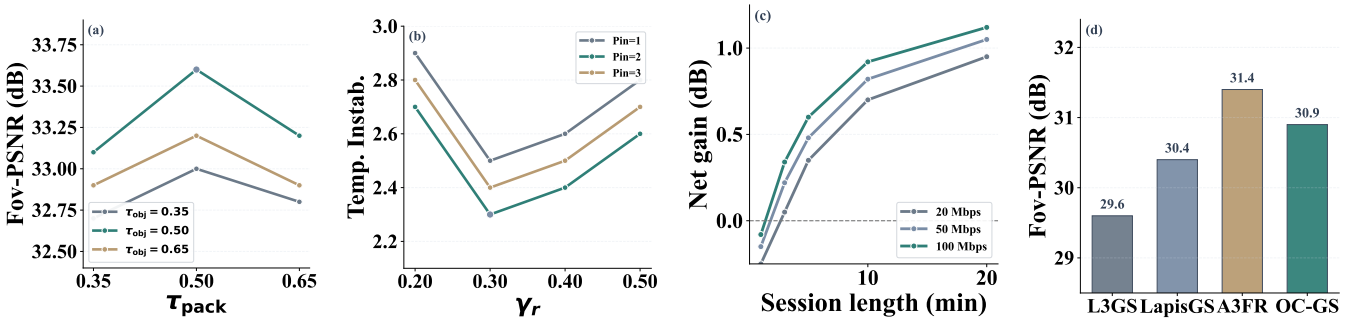

Figure 2: Supplementary sensitivity and boundary-case results. Panel (a) sweeps τ_{pack} and compares three τ_{obj} settings in Fov-PSNR. Panel (b) sweeps γ_r and compares three pinning settings in Temporal Instability. Panel (c) plots session length versus net gain under three bandwidth regimes. Panel (d) reports the Mip-NeRF360 *bonsai* failure case and compares *OC-GS* against L3GS, LapisGS, and A3FR using Fov-PSNR.

Figure 3: Additional qualitative comparison under the same setting as the main-text qualitative figure: the same event, the same budget, and a fixed 50 ms snapshot after a gaze or saccade shift. This view expands the early-recovery comparison with more scenes and method pairs while preserving the event-aligned setting used in the main paper.

Figure 4: Qualitative comparison after stabilization. Unlike Fig. 3, which emphasizes early recovery, this figure compares the same methods after the rendered view has settled, highlighting residual quality differences once transient recovery effects have largely disappeared.

again within the same session, and let B_{tot} denote the total fetched bytes in that session. We report

$$\text{Refetch} = \frac{B_{\text{re}}}{B_{\text{tot}}}.$$

Units reused from cache without an intervening eviction are not counted as refetch. Reported values are session averages.

Temporal Instability is the windowed fluctuation of attended quality on the session clock. Let $Q_t = \text{FovPSNR}_t$ and let $\mathcal{W}$ be the set of sliding windows of length $10\Delta_s$ (i.e., 1 s). We report

$$\text{TI} = \frac{1}{|\mathcal{W}|} \sum_{W \in \mathcal{W}} \sqrt{\frac{1}{|W|} \sum_{t \in W} (Q_t - \bar{Q}_W)^2}, \qquad \bar{Q}_W = \frac{1}{|W|} \sum_{t \in W} Q_t,$$

so lower values mean less attended-quality oscillation under revisits and budget coupling.

1.4 Supplementary Sensitivity and Boundary-Case Results

To complement the main quantitative results, we summarize four supplementary experiment blocks in a single four-panel diagnostic figure, following the same compact single-row presentation style as the motivation controls.

1.5 Additional Qualitative Comparisons

To complement the single qualitative case in the main paper, we include two additional visual comparisons with distinct roles. Figure 3 extends the main-text qualitative setting to more scenes and method pairs under the same event, the same budget, and the same fixed 50 ms snapshot after a gaze or saccade shift, so it should be read as an *early-recovery* comparison. Figure 4 uses the same baselines after the response has stabilized, so it should be read as a *settled-quality* comparison that isolates residual quality differences after transient recovery effects have largely disappeared.

1.5.1 Routing Operating-Point Sensitivity. This subsection reports the offline routing sensitivity analysis corresponding to Sec. 4. Panel (a) of Fig. 2 sweeps τ_{pack} while comparing several fixed τ_{obj} settings, so that the operating-point choice can be read directly as a sweep rather than inferred from mixed metrics. Alongside the selected point, we report objectization coverage ρ_{obj}, atomicity ratio ρ_{micro}, Fov-PSNR, recovery latency, and refetch ratio.

1.5.2 Scheduler Sensitivity Under Revisits. This subsection reports the online sensitivity analysis corresponding to Sec. 5. Panel (b) of Fig. 2 sweeps γ_r while comparing several pinning settings, directly showing which anti-thrashing strength gives the best trade-off under different revisit-stability regimes. We additionally report recovery latency, refetch ratio, and long-session churn around the selected setting. This experiment most directly supports the claim that stability gains arise from the full control loop rather than from predictor strength alone.

1.5.3 Cold-Start and Amortization Accounting. Table 1 reports the cold-start overhead at 100 Mbps and 20 Mbps. Panel (c) of Fig. 2 extends that summary with an amortization curve whose horizontal axis is session length and whose lines compare different bandwidth regimes. This result is secondary to routing and scheduling sensitivity, but it clarifies where the one-time semantic payload cost is recovered.

1.5.4 Bonsai Failure Case Against Baselines. This subsection presents a concrete failure case rather than a shared hard subset such as glass or diffuse content. Panel (d) of Fig. 2 focuses on the Mip-NeRF360 *bonsai* scene, whose attended region contains many small structures and fragmented boundaries within a compact field of view. In this regime, objectization becomes less clean and the scheduler receives a more fragmented executable frontier, so the advantage of object-centric packaging weakens. The resulting comparison shows a case where A3FR attains slightly higher Fov-PSNR than *OC-GS*, while *OC-GS* remains ahead of L3GS and LapisGS. The purpose is to expose a genuine boundary of the current design rather than to restate average-case gains from the main table.

References

[1] John Duncan. 1984. Selective attention and the organization of visual information. *Journal of experimental psychology: General* 113, 4 (1984), 501.
[2] Robert Egly, Jon Driver, and Robert D Rafal. 1994. Shifting visual attention between objects and locations: evidence from normal and parietal lesion subjects. *Journal of Experimental Psychology: General* 123, 2 (1994), 161.
[3] Robert D Gordon, Sarah D Vollmer, and Megan L Frankl. 2008. Object continuity and the transsaccadic representation of form. *Perception & psychophysics* 70, 4 (2008), 667–679.
[4] Susana Martinez-Conde and Stephen L Macknik. 2017. Unchanging visions: the effects and limitations of ocular stillness. *Philosophical Transactions of the Royal Society B: Biological Sciences* 372, 1718 (2017), 20160204.
[5] Susana Martinez-Conde, Stephen L Macknik, Xoana G Troncoso, and Thomas A Dyar. 2006. Microsaccades counteract visual fading during fixation. *Neuron* 49, 2 (2006), 297–305.

[6] Moaaz Hudhud Mughrabi, Aunnoy K Mutasim, Wolfgang Stuerzlinger, and Anil Ufuk Batmaz. 2022. My eyes hurt: Effects of jitter in 3d gaze tracking. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, 310–315.

[7] Anjul Patney, Joohwan Kim, Marco Salvi, Anton Kaplanyan, Chris Wyman, Nir Benty, Aaron Lefohn, and David Luebke. 2016. Perceptually-based foveated virtual reality. In *ACM SIGGRAPH 2016 Emerging Technologies* (Anaheim, California) *(SIGGRAPH '16)*. Association for Computing Machinery, New York, NY, USA, Article 17, 2 pages. doi:10.1145/2929464.2929472

[8] Miguel Fabian Romero-Rondón, Lucile Sassatelli, Frédéric Precioso, and Ramon Aparicio-Pardo. 2018. Foveated streaming of virtual reality videos *(MMSys '18)*. Association for Computing Machinery, New York, NY, USA, 494–497. doi:10.1145/3204949.3208114

[9] Jan Theeuwes, Sebastiaan Mathôt, and Alan Kingstone. 2010. Object-based eye movements: The eyes prefer to stay within the same object. *Attention, Perception, & Psychophysics* 72, 3 (2010), 597–601.

[10] Shaun P Vecera and Martha J Farah. 1994. Does visual attention select objects or locations? *Journal of Experimental Psychology: General* 123, 2 (1994), 146.